Experiment No: 04
Experiment Name: Verification of Kirchhoff's Current Law.
Course Code: EEE-1209
Course Title: Electrical Circuit lab

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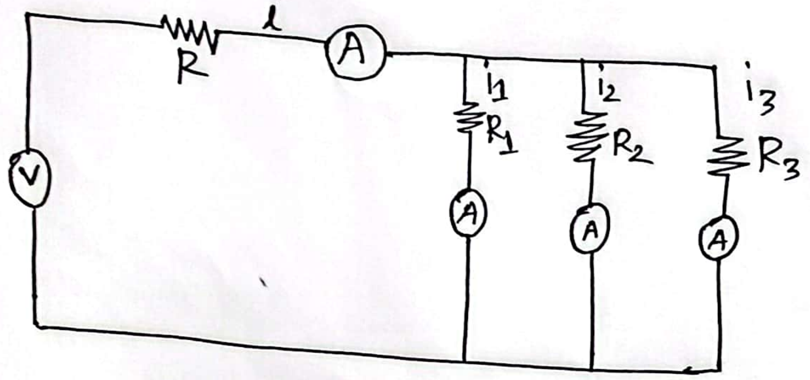
Experiment name: Verification of KCL (Kirchoff's Current Law)



Data Table:

observation	Voltage	Actual current (I)	I_1	I_2	I_3	$I_{total} = i_1 + i_2 + i_3$	Error	Error (%)
1	10V	5.9	1	4.7	0.1	5.8	0.1	1.69%
2	12V	6.9	1.1	5.5	0.1	6.7	0.2	2.89%

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Experiment name: Verification of KCL (Kirchoff's current Law)

Theory: The algebraic sum of the currents meeting at a node is equal to zero.

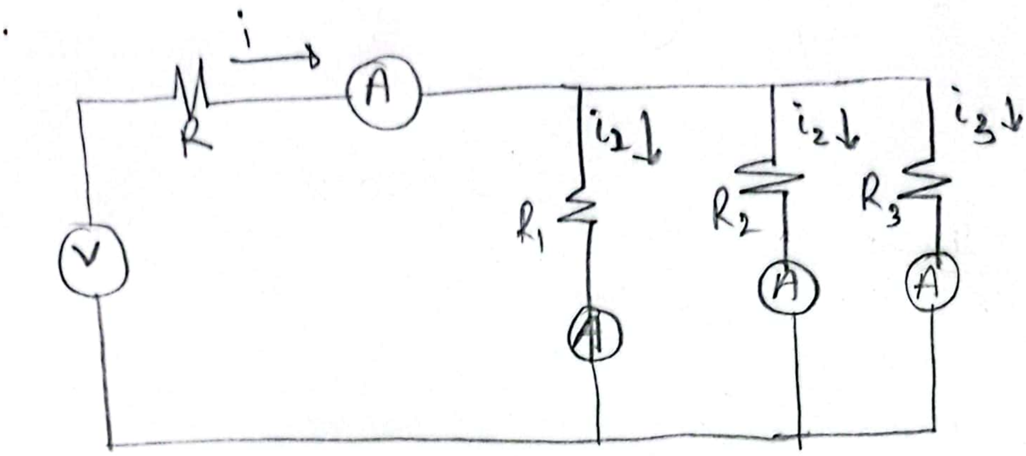
Required Equipment:

1. RPS (Reguar power supply)
   Range (0-30V) & Quantity 2
2. Resistance
   Range (1KΩ, 4.7KΩ, 3.6KΩ) &
   Quantity 4.
3. Ammeter
   Range (0-30mA) MC &
   Quantity 3.
4. Voltmeter.
   Range (0-30V) MC &
   Quantity 3.
5. Breadboard & wires.

Precautions:

1. Voltage control Knob should be kept at minimum position.
2. Current control Knob of RPS should be kept at maximum position.

Circuit Construction: current (KCL)



Procedure of KCL:

1. Give the connections as per the circuit diagram.
2. Set a particular value in RPS.
3. Note down the corresponding ammeter reading.
4. Repeat the same for different voltages.

KCL- Theoretical Values:

observation	Voltage	Actual Current (I_A) mA	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I=I_1+I_2+I_3$ (mA)	Error ($I_A \sim I$)
1	10 V	5.6	4.5	0.9	0.1	5.5	0.1
2	12 V	6.8	5.4	1.1	0.1	6.6	0.2

Result: From the data table, we find that current through different paths was different. Now we are going to test some of those data.

SL. No	I_A	$I=I_1+I_2+I_3$
1	5.6	5.5
2	6.8	6.6

From the table, we can see that $I_1+I_2+I_3$ value is always near I_A.

Conclusion:

From the result section, we can see that current entering and exiting were the same. The algebraic sum of all currents entering and exiting a node was zero. From this, we can say that kirchoff's current law is true in real world.